

orbit_demo verification summary

Package section 07_verification of orbit_demo_design_review, written 2026-10-06. Links are relative to this file. Paths in backticks without a link (reports/..., build/..., pd/..., ORFS image paths) are repository, build or tool-image paths. Byte-identical copies of the repository directories pd/, mk/, fault/, formal/, synth/, rtl/ and of part of scripts/ and docs/ are in [04_digital_design/source/](#) (cmp 2026-10-06); copies of reports/{formal,fault,synth,ecc,mutation} and of most of reports/pdsep/ are in this folder (section 2). build/... and ORFS image paths are not in the package unless section 2 lists a copy. Every result row carries one status label: VERIFIED (a tool log or report shows it), REPRODUCED 2026-10-06 (re-run for this package), TARGET (design goal, not measured), ASSUMPTION, CLAIM (unverified) (stated in a document, no log found), MISSING, N/A. Discrepancies are collected in the discrepancy register ([09_review_notes](#)); this document only flags them.

Scope: the reviewed layout is the copy-separated sky130hd run (variant sep, 7.2 ns). The baseline sky130hd run (variant base, 7.0 ns) is used only for comparison and is labelled as such. The ECC block is standalone and not in the layout (section 15).

1 Design revision covered by this package

Package revision table, verbatim from the package revision record:

Item	Revision / identifier	Note
Design	orbit_demo (ORBIT-AI 4-lane INT8 digital demonstrator)	top module orbit_demo, LANES=4
RTL	rtl/orbit_demo.v, orbit_mac_lane.v, orbit_thermal_tmr.v, orbit_keep_reg.v at git commit 26566f0 (2026-09-29 18:54 UTC); unchanged through the package commit	sha256 a3fffb22... / efef3d33... / bba77a5a... / ea05bdc1... (full hashes in MANIFEST.sha256 at the package root)
Specification	docs/SPEC.md at 26566f0 (never revised)	sha256 aef5d9fb...
Concept brief	docs/orbit-ai-design-brief.pdf, "ORBIT-AI v0.1, 29 September 2026" (never revised; describes an earlier project state, see discrepancy register)	sha256 9d5f33d3...
Reviewed layout	ORFS sky130hd, variant sep (copy-separation fences), run 2026-09-29 21:56-22:12 UTC, outputs build/pd_sep/results/sky130hd/orbit_demo/sep/	6_final.gds sha256 cd19afa9...; 6_final.def f5c544f3...; 6_final.v 68093c41...; 6_final.spef 014e655a...
Layout configuration	pd/sky130hd_sep/config.mk and regions.tcl at c10c59b; constraint.sdc at 21917cf (set clk_period 7.2)	The run (21:56 UTC) used config.mk/regions.tcl as in ed553f8, identical to c10c59b once comments are stripped. It used the working-tree constraint.sdc with period 7.2 ns, which was committed in 21917cf at 22:00 UTC and is byte-identical to the flow copy build/pd_sep/sdc/sky130hd/sep.sdc. ed553f8's constraint.sdc has a 7.0 ns period, so a rebuild must not use it
Flow / tools	ORFS docker image tag openroad/orfs:latest; the only local image is sha256:2e5bf6fe865e... (created 2026-09-29 02:00 UTC, before the run). Linking the run to this digest is an ASSUMPTION: the run logged no digest. OpenROAD prints version "unknown". KLayout 0.30.12 (DRC/LVS). Yosys 0.69+154 (local sims/schematics)	No PDK commit is recorded in the logs
Process / library	SkyWater SKY130, sky130_fd_sc_hd, liberty sky130_fd_sc_hd_tt_025C_1v80 (the single corner ORFS optimised at)	—
Package assembled	2026-10-06; design inputs copied from the repository tree at commit 0495cfa (branch claudes/hopeful-rubin-0yf8io)	Every later commit on the branch changes only review/ or is empty (git diff --name-only 0495cfa..HEAD outside review/ is empty). Re-runs made for this package are dated 2026-10-06

Other implementations in the repo (sky130hd baseline at 7.0 ns, variant base; IHP SG13G2) are reference runs only and are NOT the reviewed layout. Checked for this section on 2026-10-06: MANIFEST.sha256 is at the package root ([../MANIFEST.sha256](#), generated last on 2026-10-06; register D-53). The fault summary [fault/summary.md](#) lines 15-20 records the full sha256 of the four RTL files, and they match the table prefixes. The SPEF copy [6_final.spef.gz](#) decompresses to sha256 014e655ae938245a..., the value in the table. git diff --name-only 0495cfa..HEAD outside review/ is empty (checked 2026-10-06 at HEAD 23c72ac); no design, report or flow file changed after the package commit.

2 Evidence in this folder

Unless a source cell names another path, the original of a package file is the folder origin below plus the same file name.

Package folder	Contents	Origin (repository or build path)	How it got here
orfs_logs/	ORFS step logs and metrics JSON of the sep run	build/pd_sep/logs/sky130hd/orbit_demo/sep/	copied before this section; 39 of 40 files, byte-identical (cmp 2026-10-06); the 40th, 5_1_grt.json, is in additional_evidence/flow/
orfs_reports/	6_finish.rpt, 6_drc.lyrdb, 6_drc_count.rpt, synth_stat/check, stage timing reports	build/pd_sep/reports/sky130hd/orbit_demo/sep/	copied before this section; with 06_physical_design/orfs_images/ 23 of 25 files, byte-identical; the other two (drt_antennas.log, grt_antennas.log) are in additional_evidence/flow/
published_reports/	results.md, checks.json, storage_audit.txt, power, paths, separation	reports/pdsep/	copied before this section; 18 of 19 files, byte-identical
gls/	post-route GLS log and injection list	build/pd_sep/gls/sep/	copied before this section; byte-identical
formal_rerun_2026-10-06/	make formal (12 tasks) and make formal-vacuity re-run	review scratch build	re-run 2026-10-06
sta_corners_sep_2026-10-06/	multi-corner STA of the sep layout, scripts and logs	review scratch	new run 2026-10-06
formal/	summary.md, vacuity.md, logs/ (13 tasks, 19 vacuity mutants)	reports/formal/	copied by this section (cp, not moved)
fault/	summary.md, campaign.md, campaign_per_bit.tsv, injection/, logs/	reports/fault/	copied by this section
synth/	generic synthesis, audits, negative control, equivalence, GLS	reports/synth/	copied by this section
ecc/	results.txt, synth_stat.txt, logs/	reports/ecc/	copied by this section
mutation/	summary.md, results.json, tb_mutation_gaps.v	reports/mutation/	copied by this section

Package folder	Contents	Origin (repository or build path)	How it got here
baseline_reference/	sta_corners_baseline_7p0ns.txt; negctl_cfgkeep_audit.txt; negctl_nokeep_audit.txt (baseline ORFS keep controls); results_baseline_7p0ns.md	reports/pd/sky130hd/: sta_corners.txt, negctl_*_audit.txt, results.md	copied by this section; sta_corners.txt and results.md renamed
lec_inputs/	4_rsz_lec_test.yml, 6_final_lec_test.yml (kepler-formal test configurations; they name the compared netlists, which are in additional_evidence/lec_netlists/)	build/pd_sep/objects/sky130hd/orbit_demo/sep/	copied by this section
lvs/	orbit_demo_extracted.cir.gz (layout-extracted netlist from KLayout LVS)	build/pd_sep/results/ sky130hd/orbit_demo/sep/	gzip copy of orbit_demo_extracted.cir by this section (decompresses byte-identical)
rerun_2026-10-06/	re-run logs: make_fault, make_formal-extra, a second make_formal-vacuity, make_synth, make_ecc, ECC_sec_bmc sanity, sep_storage_audit with the current script, final STA/power/IR/EM query, GLS re-run, GLS netlist-mutant control	review_scratch_review_work/ (formal_fault, synth, ecc, pd_sep, pd_ref)	copied by this section; logs contain scratch paths
additional_evidence/	lec_netlists/: 1_synth_lec.v, 4_before_rsz_lec.v, 4_after_rsz_lec.v, 6_final_lec.v (the netlists kepler-formal compared). flow/: make_pd_sep.out (console log of the reviewed run, 7931 lines), 5_1_grt.json, drt_antennas.log and grt_antennas.log (0 bytes), 6_lvs.lvsdb (KLayout LVS database, 17.7 MB uncompressed), 6_final_concat.cdl (LVS reference netlist)	build/pd_sep/results/sky130hd/orbit_demo/sep/ (*_lec.v, 6_lvs.lvsdb); build/pd_sep/make_pd_sep.out; build/pd_sep/logs/sky130hd/orbit_demo/sep/5_1_grt.json; build/pd_sep/reports/sky130hd/orbit_demo/sep/ (antenna logs); build/pd_sep/objects/sky130hd/orbit_demo/sep/6_final_concat.cdl	added at the final step, 2026-10-06; gzip copies (the two 0-byte logs as is), decompress byte-identical (cmp)

Not copied (size): formal and fault counterexample/cover VCDs under build/formal/ and build/fault/, per-trial campaign log build/fault/campaign/trials.tsv. Also not copied: reports/pdsep/orbit_demo_sky130hd_sep.gds.gz (decompresses to the same sha256 cd19afa9... as [6_final.gds.gz](#)). Added at the final step (2026-10-06; register D-99), see [additional_evidence/](#) above:
build/pd_sep/results/sky130hd/orbit_demo/sep/6_lvs.lvsdb (17.7 MB), build/pd_sep/make_pd_sep.out (console log, 7931 lines), 5_1_grt.json (global-route metrics, 586912 B) from the ORFS log folder, drt_antennas.log and grt_antennas.log (both 0 bytes) from the ORFS report folder, 6_final_concat.cdl, and the LEC netlists 1_synth_lec.v, 6_final_lec.v, 4_before_rsz_lec.v, 4_after_rsz_lec.v from build/pd_sep/results/sky130hd/orbit_demo/sep/. With these and [lec_inputs/](#) the LEC can be re-run from the package; the yml files name the build paths, and the TT liberty and kepler-formal come from the ORFS image. 6_final.odb is in [06_physical_design/layout_db/](#).

3 Tools, conditions, targets and assumptions

3.1 Tools

Tool	Version as recorded	Used for	Source	Status
OpenROAD (OpenSTA, TritonRoute, TritonCTS, OpenRCX, PSM, ANT)	prints "unknown"; build features -GPU +GUI -Python	P&R, STA, extraction, IR, antenna	6_report.log line 1; SPEF header *VERSION "unknown"	VERIFIED
ORFS docker image	tag openroad/orfs:latest; local image digest sha256:2e5bf6fe865e..., created 2026-09-29 02:00 UTC	whole flow	no log records the digest; only one ORFS image on the host	ASSUMPTION (that this image ran the 2026-09-29 flow)
ORFS commit, PDK (open_pdk / skywater-pdk) commit	not recorded	—	—	MISSING
Yosys (ORFS synthesis)	0.68+post (git sha1 UNKNOWN)	sky130hd mapping	1_2_yosys.log line 1	VERIFIED
KLayout	0.30.12	DRC, LVS, GDS merge	6_drc.log line 1; 6_lvs.log line 1	VERIFIED
kepler-formal	version not printed; tool name only in ORFS flow/scripts/formal_check.tcl and pd/collect_pd.py line 150	LEC	6_final_lec_check.log	MISSING (version)
Yosys / SBY / Yices / bitwuzla (formal, fault, generic synthesis, ECC)	Yosys 0.69+154 (git sha1 30d62572e-dirty); SBY v0.69; Yices 2.7.0; bitwuzla 0.9.1; abc pdr and aiger suprove from the same OSS CAD Suite, versions not stated	formal, fault, formal, synth, ECC	formal/summary.md line 4; fault/summary.md line 11	VERIFIED
Icarus Verilog	14.0 (devel) (s20260301-500-g2e81fccb-dirty)	SEU campaign, GLS, ECC sim	fault/summary.md line 11	VERIFIED
Verilator	5.053 devel rev v5.052-233-gf5f9ddef9	ECC lint	lint_verilator.log line 1	VERIFIED

3.2 Common timing / physical conditions

Unless stated: sep layout 6_final.odb / [6_final.sdc](#) / [6_final.spef.gz](#); create_clock clk -period 7.2000 (6_final.sdc line 8); set_propagated_clock (line 9); set_input_delay 1.4400 on 79 inputs and set_output_delay 1.4400 on 135 outputs (0.2 x period); no set_clock_uncertainty, set_timing_derate, set_load, set_driving_cell or set_input_transition in 6_final.sdc (grep 2026-10-06). Liberty sky130_fd_sc_hd__tt_025C_1v80.lib (TT, 25 C, 1.80 V), the only liberty ORFS read ([6_report.log](#) line 6).

3.3 Targets and assumptions referenced in this section (not results)

Item	Value	Source	Status
Demonstrator clock requirement	none in SPEC.md; 7.2 ns is the shortest period tried that closed at TT (period_exploration.md)	SPEC.md	MISSING (requirement)
Concept chip: 16 tiles, 200-800 MHz, 6.55-26.21 TOPS	concept targets for a different, unbuilt chip (p.1 "unvalidated targets"; p.5 also calls the tile count and clock range project assumptions); they do not apply to orbit_demo and nothing in this section verifies them	brief p.1-2, p.5	TARGET

Item	Value	Source	Status
Concept chip: 40 W per chip	conceptual sizing allocation, not a power estimate (p.1 "sizing assumption - not an estimate"; p.2 "for conceptual sizing only"); does not apply to orbit_demo	brief p.1-2, p.5	ASSUMPTION
SPEC section 5 integrity target	an upset in one copy of accumulator/result storage never transfers a wrong out_data; it causes a stop	SPEC.md lines 93-94	TARGET (checked in section 12)
I/O delay 20 % of period	placeholder for the unknown outside world	constraint.sdc lines 19-27	ASSUMPTION
Fault model	single upset per trace/trial; synchronous model, timing closure assumed, metastability out of scope	fault/summary.md lines 33-41	ASSUMPTION
Copy-separation acceptance limits	same-bit centre >= 20 um, edge gap >= 10 um; no rationale given	scripts/pdsep_separation.py lines 46-48	TARGET

4 Physical verification

4.1 DRC

Check	Result	Tool + version	Conditions	Status	Source
Detailed-route DRC	0 violations after the last iteration (35 at the preceding report step); 5_route_drc.rpt is empty (0 bytes)	TritonRoute in OpenROAD "unknown"	detailed_route -droute_end_iter 64; router rules from the sky130hd tech LEF. DRT-0349: LEF58_ENCLOSURE without CUTCLASS not supported, enclosure rules skipped for mcon, via, via2, via3, via4	VERIFIED	5_2_route.log lines 10-21, 973, 1007; 5_route_drc.rpt
KLayout DRC, ORFS deck	0 markers: 6_drc_count.rpt = 0; 6_drc.lyrdb has 145 rule categories and 0 items	KLayout 0.30.12, ORFS deck sky130hd.lydrc ("v1.0 : initial release", 2020-10-04)	Deck scope: FEOL = false, BEOL = true, OFFGRID = true (deck lines 46-48, read from the ORFS image). Log runs only "DRC section", "BEOL section", "OFFGRID-ANGLES section". Categories: li, ct, m1-m5, via-via4, nsm, pad.2, plus *_OFFGRID / *_angle. No FEOL rules (well, diff/tap, poly, licon, implants), no metal-density rules, no antenna or latch-up rules. Run on 6_final.gds without metal fill. This is NOT a sign-off DRC	VERIFIED	6_drc_count.rpt ; 6_drc.lyrdb ; 6_drc.log lines 1, 257-258, 1222; command in build/pd_sep/make_pd_sep.out lines 5517-5519
Metal fill	not inserted: step 6_1_fill only copies 5_route.odb to 6_1_fill.odb	OpenROAD "unknown"	ORFS sky130hd platform default	VERIFIED (that no fill was run)	6_1_fill.log line 10
Metal density on a filled GDS	not run	—	needs fill and a density check. The tech LEF read by the flow defines MAXIMUMDENSITY 70 / DENSITYCHECKWINDOW 700 700 for met1-met4 (no minimum density); no step compared the layout with it	MISSING	sky130_fd_sc_hd.tlef (ORFS image copy) lines 120-121, 165-166, 206-207, 247-248; read by the flow per 1_synth.log line 7 (ODB-0227)
FEOL DRC (full sky130 rule deck, e.g. open_pdk KLayout sign-off deck)	not run	—	—	MISSING	—
Magic DRC	not run; no Magic log or tech file use in the flow	—	—	MISSING	—

The PASS row in [results.md](#) line 20 ("KLayout DRC PASS, 0 markers") does not state the deck scope; see discrepancy register (09_review_notes).

4.2 LVS

Check	Result	Tool + version	Conditions	Status	Source
Layout vs schematic	"INFO : Congratulations! Netlists match."	KLayout 0.30.12, ORFS deck sky130hd.lylvs	Layout: 6_final.gds extracted at device level (orbit_demo_extracted.cir). Reference: 6_final_concat.cdl = 6_final.cdl (written by OpenROAD from 6_final.odb) concatenated with the platform cell CDL sky130hd.cdl. The reference comes from the same database as the GDS, so this confirms stream-out and cell connectivity, not an independent netlist	VERIFIED	6_lvs.log lines 1, 649, 652; 6_cdl.log ; reference 6_final.cdl.gz ; extracted orbit_demo_extracted.cir.gz ; see note below
Independent LVS (Magic extraction + Netgen against 6_final.v or the synthesized netlist)	not run	—	—	MISSING	—

Note: the concatenation command is in build/pd_sep/make_pd_sep.out line 7244 ([make_pd_sep.out.gz](#)); the reference netlist is [6_final_concat.cdl.gz](#); the LVS database is build/pd_sep/results/sky130hd/orbit_demo/sep/6_lvs.lvldb (17.7 MB), packaged as [6_lvs.lvldb.gz](#). Deck paths are inside the ORFS image under /OpenROAD-flow-scripts/flow/.

4.3 Logic equivalence (LEC) inside the flow

Check	Result	Tool + version	Conditions	Status	Source
4_rsz LEC, published as "LEC synth vs post-CTS repair"	"Circuits are IDENTICAL"; 613 / 613 PIs, 1177 POs, SAT UNSAT	kepler-formal (version MISSING), called by ORFS cts.tcl	Compares 4_before_rsz_lec.v with 4_after_rsz_lec.v: the netlists immediately before and after repair_timing inside the CTS step (ORFS flow/scripts/cts.tcl lines 61-71). It is NOT synthesis vs post-CTS. Liberty TT	VERIFIED	4_rsz_lec_check.log lines 4, 6, 95; 4_rsz_lec_test.yml ; 4_1_cts.log line 469; compared netlists now included: 4_before_rsz_lec.v.gz , 4_after_rsz_lec.v.gz
6_final LEC, "synth vs final"	"Circuits are IDENTICAL"; 603 / 613 PIs, 1177 common POs, SAT UNSAT	kepler-formal (version MISSING)	Compares 1_synth_lec.v with 6_final_lec.v. Combinational miter with the 521 dfxtp_1 Q pins as cut points; 1177 POs = 521 CLK + 521 D + 135 output bits. The 10 extra PIs are tie-cell outputs (conb_1 HI/LO: 1+1 in synthesis, 6+6 in the final netlist). Does not cover RTL vs synthesis	VERIFIED	6_final_lec_check.log lines 4, 6, 8-24, 44, 109; 6_final_lec_test.yml ; compared netlists now included: 1_synth_lec.v.gz , 6_final_lec.v.gz

The label in [results.md](#) line 22 and [pd/collect_pd.py](#) line 151 misnames the 4_rsz check; see discrepancy register (09_review_notes). RTL-to-gate equivalence is in section 13.

4.4 Antenna

Check	Result	Tool + version	Conditions	Status	Source
Global-route antenna repair	31 antenna violations found, 0 after repair	OpenROAD GRT / ANT, "unknown"	ORFS default antenna rules from the sky130hd tech LEF	VERIFIED	5_1_grt.log lines 663-668
Detailed-route antenna repair	18 diodes inserted, then 2 more	OpenROAD DRT / GRT-0015	—	VERIFIED	5_2_route.log lines 425, 765
Final antenna check	0 net violations, 0 pin violations; 20 antenna cells in the final cell report	OpenROAD ANT-0002 / ANT-0001	routed database	VERIFIED	5_2_route.log lines 1056-1059; 6_report.json line 14; 6_report.log cell report
Antenna check by KLayout or Magic	not run: the KLayout deck has no antenna rule categories	—	—	MISSING	6_drc.lyrdb category list

5 Extracted-parasitic results

Check	Result	Tool + version	Conditions	Status	Source
Parasitic extraction	6886 nets, 30095 rsegs, 30095 caps, 60471 coupling caps; written to 6_final.spef (6827076 bytes, sha256 014e655a...) and read back for the final STA, power and IR reports	OpenRCX in OpenROAD "unknown"	Rules <code>rcx_patterns.rules</code> of the ORFS sky130hd platform (the only RCX rules file there); one process corner X, <code>ext_model_index 0</code> (a single nominal extraction corner); coupling threshold 0.1 fF; <code>max_merge_res 50.0</code> . Provenance and qualification of <code>rcx_patterns.rules</code> are not documented	VERIFIED	6_report.log lines 16-26; 6_final.spef.gz header; ORFS <code>flow/scripts/final_outputs.tcl</code> lines 27-37
STA on the SPEF	final ORFS STA, power and IR use the OpenRCX SPEF; the 2026-10-06 corner runs read the same SPEF (<code>read_spef</code>) at every liberty corner	OpenSTA in OpenROAD "unknown"	see section 6	VERIFIED	ORFS <code>final_outputs.tcl</code> line 37; sta_corners_extra.tcl ; pd/sta_corners.tcl lines 14-18
RC corners (min/max extraction rules)	not available; nominal SPEF used at every PVT corner	—	—	MISSING	pd/sta_corners.tcl lines 9-10
Post-layout SPICE / transistor-level simulation	not run (the LVS-extracted device netlist exists but carries no parasitics and was never simulated)	—	—	MISSING	—
Post-layout timing (SDF-annotated) gate-level simulation	not run; every GLS in the package is zero-delay	—	—	MISSING	tb_gls.v header lines 12-13

6 Timing

6.1 ORFS final STA at TT (the flow's own final report; single corner)

Check	Result	Tool + version	Conditions	Status	Source
Setup	WNS +0.0310081 ns, TNS 0, 0 setup violations	OpenSTA in OpenROAD "unknown"	section 3.2; TT 25 C 1.80 V; OpenRCX SPEF	VERIFIED	6_report.json lines 30, 32; 6_finish.rpt lines 5, 15, 416
Worst setup path	<code>in_a[27]</code> (input, 1.44 ns external delay) -> <code>g_lane[3].u_lane.u_res_a/q[29]\$_SDFFE_PN0P_</code> , slack 0.03 ns (MET); an I/O path that crosses the die from the east-edge pin to the copy-A fence	same	same	VERIFIED	worst_setup_path.txt lines 1-2, 125; 6_final.def.gz : pin <code>in_a[27]</code> PLACED (345895 109140) dbu, flip-flop PLACED (43700 108800) dbu (0.001 um/dbu)
Hold	WNS +0.435959 ns, TNS 0, 0 hold violations; worst path <code>u_thermal.phase\$_SDFFE_PN0_</code> -> itself, slack 0.44	same	same	VERIFIED	6_report.json lines 31, 33; worst_hold_path.txt line 53; 6_finish.rpt line 421
Clock skew	setup skew 0.0538434 ns (latency 0.65 at <code>u_thermal.u_copy0/q[0]\$_SDFFE_PN0P_/CLK</code> , -0.59 at <code>g_lane[3].u_lane.u_res_b/q[20]\$_SDFFE_PN0P_/CLK</code>); one TritonCTS H-tree for all 521 sinks, 56 clock buffers	OpenSTA / TritonCTS	propagated clock	VERIFIED	6_report.json line 36; 6_finish.rpt lines 23-30; 4_1_cts.log lines 17-43
Min period / fmax incl. I/O paths	7.17 ns / 139.49 MHz (<code>report_clock_min_period - include_port_paths</code> , with the assumed 1.44 ns I/O delays held fixed)	OpenSTA	same	VERIFIED	6_finish.rpt line 20; ORFS <code>report_metrics.tcl</code> line 39
Min period reg-to-reg (no port paths)	5.85 ns / 170.82 MHz; same-database re-query also returns setup WNS 0.031008 ns, hold WNS 0.435959 ns, 7.17 ns with port paths	OpenSTA in OpenROAD "unknown", image sha256:2e5bf6fe865e	sta.tcl : read 6_final.odb/sdc , setRC.tcl , 6_final.spef	REPRODUCED 2026-10-06	sta.log lines 5, 10-11
Intermediate (estimate-based) timing	did not close before the final RCX-based STA: RSZ-0062 Unable to repair all setup violations at floorplan, CTS and global route	OpenROAD resizer	placement/route estimates	VERIFIED	2_1_floorplan.log line 362; 4_1_cts.log line 463; 5_1_grt.log line 620

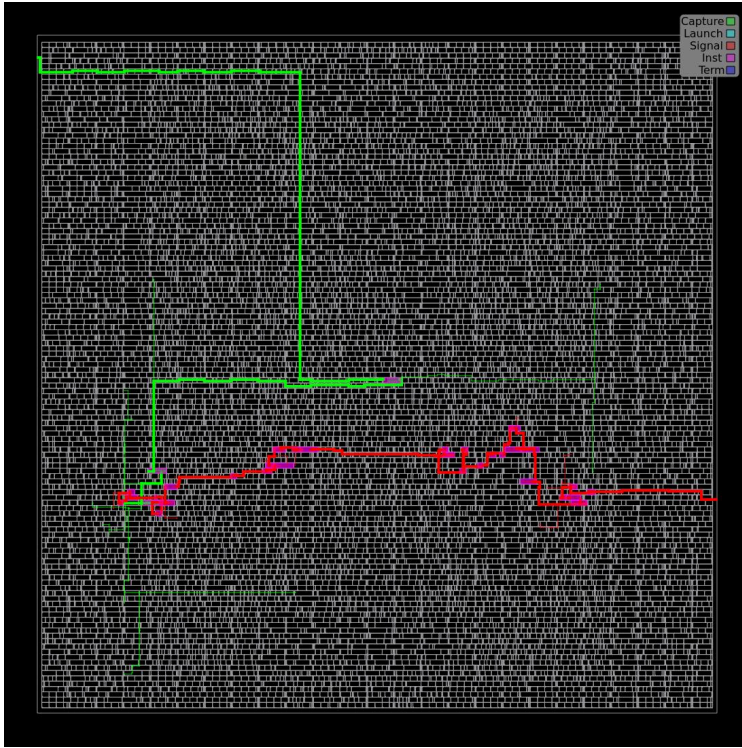


Figure 1: ORFS GUI export `final_worst_path` of the sep layout (TT, 7.2 ns): the worst setup path runs from the east-edge port `in_a[27]` to the copy-A fence at the west edge. Source [final_worst_path.webp.png](#) (= `build/pd_sep/reports/sky130hd/orbit_demo/sep/final_worst_path.webp.png`).

6.2 Multi-corner re-time of the sep layout (new, 2026-10-06)

Conditions for every row: `pd/sta_corners.tcl` unmodified, run through `scripts/run_pd.sh --shell` with `openroad -no_init -threads 1` in image `openroad/orfs:latest` (sha256:2e5bf6fe865e), inputs `build/pd_sep/results/sky130hd/orbit_demo/sep/6_final.{odb, sdc, spef}`, 7.2000 ns, 1.44 ns I/O delay, propagated clock, nominal OpenRCX SPEF at every corner. The design was optimised by ORFS at TT only; SS and FF re-time the same routed database (information only). SS/FF liberty from the master branch of `efabless/skywater-pdk-libs-sky130_fd_sc_hd` (sha256 prefixes `ss 9b24f0db3967ac67`, `ff fb61d91c55a7f85b`; no commit pinned), reused from `build/pd/platform/sky130hd/corners/`. The script header line "Constraint: the committed SDC" is fixed text; the input is the ORFS-written `6_final.sdc` (same 7.2 ns); see discrepancy register (09_review_notes).

Corner (liberty)	Setup WNS / TNS (ns)	Hold WNS / TNS (ns)	Min period / fmax (incl. I/O)	Worst setup path	Tool + version	Status	Source
tt (tt_025C_1v80)	+0.031 / 0.000	+0.436 / 0.000	7.17 ns / 139.49 MHz	<code>in_a[27] -> g_lane[3].u_lane.u_res_a/q[29]\$_SDFFE_PN0P_</code>	OpenSTA, OpenROAD "unknown"	REPRODUCED 2026-10-06 (matches 6.1)	sta_corners_sep.txt line 13; sta_tt.log
ss_100C_1v60 (ss_100C_1v60)	-6.131 / -2463.532: FAILS at 7.2 ns	+0.891 / 0.000	13.33 ns / 75.01 MHz	<code>in_a[12] -> g_lane[1].u_lane.u_acc_a/q[31]\$_SDFFE_PN0P_</code>	same	REPRODUCED 2026-10-06	line 14; sta_ss_100C_1v60.log lines 6-11
ff_n40C_1v95 (ff_n40C_1v95)	+2.276 / 0.000	+0.281 / 0.000 (smallest hold margin)	4.92 ns / 203.07 MHz	<code>in_b[30] -> g_lane[3].u_lane.u_res_a/q[29]\$_SDFFE_PN0P_</code>	same	REPRODUCED 2026-10-06	line 15; sta_ff_n40C_1v95.log

Supplementary queries ([sta_corners_extra.tcl](#), same setup as `pd/sta_corners.tcl`; [sta_corners_sep_supplement.txt](#), generated by `extract_supplement.py`):

Check	Result	Tool + version	Conditions	Status	Source
Worst setup slack by path class, TT	<code>reg2reg +1.346 (g_lane[0].u_lane.u_res_b/q[30]), in2reg +0.031, reg2out +0.654 (in_ready), in2out +2.577</code>	OpenSTA	<code>report_checks -from/-to class, -format end</code>	REPRODUCED 2026-10-06	supplement, corner tt
Worst setup slack by path class, SS	<code>reg2reg -4.420, in2reg -6.131, reg2out -4.142 (in_ready), in2out +1.195</code> . The SS failure is not only the I/O budget: <code>reg-to-reg</code> fails by 4.420 ns	OpenSTA	same	REPRODUCED 2026-10-06	supplement, corner ss_100C_1v60
Worst setup slack by path class, FF	<code>reg2reg +3.571, in2reg +2.276, reg2out +2.558, in2out +3.165</code>	OpenSTA	same	REPRODUCED 2026-10-06	supplement, corner ff_n40C_1v95
Setup-violating endpoints at SS	516: all 32 bits of each of the 16 registers <code>g_lane[0..3].u_lane.u_{acc_a, acc_b, res_a, res_b}</code> (512), <code>out_valid_q\$_SDFFE_PP0P_/D</code> , <code>fault_q\$_SDFFE_PP0P_/D</code> , outputs <code>in_ready</code> and <code>out_valid</code> . 0 at TT and FF. Hold-violating endpoints: 0 at all three corners	OpenSTA	<code>report_checks -slack_max 0 -group_path_count 1000000 -endpoint_path_count 1</code>	REPRODUCED 2026-10-06	sta_extra_ss_100C_1v60.log section "setup violating endpoints"

Check	Result	Tool + version	Conditions	Status	Source
Max slew / cap at SS	37 max-slew violating pins (worst _6342_/Y 2.004 ns vs limit 1.492 ns) and 12 max-capacitance violating pins (worst _4662_/Y 0.128 pF vs limit 0.115 pF). None at TT or FF	OpenSTA	report_check_types - max_slew - max_capacitance - max_fanout -violators; limits from each corner liberty	REPRODUCED 2026-10-06	sta_extra_ss_100C_1v60.log section "drv"

6.3 Baseline layout corners, for comparison only (not the reviewed layout)

Baseline variant base (no fences), build/pd/results/sky130hd/orbit_demo/base/6_final.*, 7.0000 ns, same script and liberty files, run 2026-09-29 20:15.

Corner	Setup WNS / TNS (ns)	Hold WNS (ns)	Min period / fmax	sep minus base min period	Status	Source
tt	+0.066 / 0.000	+0.437	6.93 ns / 144.21 MHz	+0.24 ns	VERIFIED	sta_corners_baseline_7p0ns.txt line 11 (= reports/pd/sky130hd/sta_corners.txt)
ss_100C_1v60	-6.016 / -2240.224	+0.893	13.02 ns / 76.83 MHz	+0.31 ns	VERIFIED	line 12
ff_n40C_1v95	+2.256 / 0.000	+0.280	4.74 ns / 210.80 MHz	+0.18 ns	VERIFIED	line 13

The committed corner report covers only the baseline layout and its header does not name the variant; read as evidence for the sep layout it understates the SS deficit; see discrepancy register (09_review_notes).

6.4 Not modelled in any timing result

Item	State	Status
On-chip variation, timing derates, clock uncertainty / jitter	none in 6_final.sdc or pd/sta_corners.tcl	MISSING
I/O environment (load, drive, input transition, real I/O budget)	not constrained; the I/O delay is the 20 % placeholder listed in 3.3	MISSING
RC corners	nominal OpenRCX SPEF at every corner	MISSING
Other liberty corners (e.g. ss_n40C_1v60, ff_100C_1v95, low-voltage SS)	only tt_025C_1v80 (ORFS image) and ss_100C_1v60, ff_n40C_1v95 (downloaded) exist on disk (search 2026-10-06)	MISSING
Multi-corner optimisation / closure at SS	ORFS optimised at TT only; SS fails at 7.2 ns	MISSING
Repo make target for corners of the sep layout	mk/pd.mk pd-corners runs only build/pd/.../base; mk/pdsep.mk has none; the 2026-10-06 run used the same commands by hand (run_corners_sep.sh)	MISSING

7 Max slew, max capacitance, max fanout

Check	Result	Tool + version	Conditions	Status	Source
Max slew, TT	0 violations; worst slack 0.326 ns against a 1.486 ns limit	OpenSTA, OpenROAD "unknown"	TT, final STA; repair_design -cap_margin 20 -slew_margin 20 applied during the flow (SLEW_MARGIN = CAP_MARGIN = 20)	VERIFIED	6_finish.rpt lines 359-371, 401; 3_4_place_resized.log line 12; config.mk lines 70-71
Max capacitance, TT	0 violations; worst slack 0.00733 pF against a 0.02107 pF limit	same	same	VERIFIED	6_finish.rpt lines 384-396, 411
Max fanout, TT	"0 violations" reported, but no fanout limit is defined (max_fanout_check_limit 1.0000000150474662e+30; finish_timing_drv_max_fanout_limit 0 in the metrics), so the PASS is vacuous	same	same	VERIFIED	6_finish.rpt lines 374-381, 406; 6_report.json line 42
Max fanout against a defined limit	not checked: no limit in the liberty, SDC or flow config	—	—	MISSING	—
Max slew / cap at SS and FF	SS: 37 slew and 12 cap violating pins; FF: none	OpenSTA	section 6.2	REPRODUCED 2026-10-06	section 6.2

The published "max slew/cap/fanout PASS" ([results.md](#) line 17) does not name the corner and counts a fanout check that has no limit; see discrepancy register (09_review_notes). The metrics keys finish_timing_drv_max_slew_limit (0.219284) and ..._max_cap_limit (0.347765) in [6_report.json](#) are the slack-to-limit ratios printed in [6_finish.rpt](#) as max_slew_check_slack_limit 0.2193 and max_capacitance_check_slack_limit 0.3478, not limits.

8 Power

Check	Result	Tool + version	Conditions	Status	Source
Total power	52.1092 mW: internal 26.3633 mW, switching 25.7459 mW, leakage 0.0279 uW (2.79317e-08 W). By group: Sequential 5.96 mW (11.4 %), Combinational 43.2 mW (83.0 %), Clock 2.90 mW (5.6 %), Macro 0, Pad 0	OpenSTA report_power, OpenROAD "unknown"	TT 25 C 1.80 V, 7.2 ns, OpenRCX SPEF; the tool's default switching activity, no VCD/SAIF annotated; the numeric default activity is not reported. A tool estimate, not a measurement	VERIFIED	6_report.json line 49 and power keys; 6_finish.rpt lines 549-561; power_default_activity.txt ; results.md lines 53-56
Same query on the same database	5.21e-02 W total, identical group table	same	sta.tcl	REPRODUCED 2026-10-06	sta.log
Activity-based power (workload VCD/SAIF), power at SS/FF	not run	—	—	MISSING	—
Baseline layout, for comparison	47.4874 mW, same TT / default-activity conditions at 7.0 ns	OpenSTA	baseline	VERIFIED	results_baseline_7p0ns.md line 53 (= reports/pd/sky130hd/results.md)

9 IR drop

Check	Result	Tool + version	Conditions	Status	Source
Static IR drop, VDD	worst 0.4539 mV (4.53888e-04 V), average 0.0958 mV, 0.03 %; "All shapes on net VDD are connected"	OpenROAD PSM analyze_power_grid (called by ORFS final_outputs.tcl lines 39-56), OpenROAD "unknown"	supply 1.80 V (PWR_NETS_VOLTAGES), corner default, total power 5.21e-02 W from the same default-activity estimate as section 8; PSM default voltage-source placement; OpenRCX SPEF	VERIFIED	6_report.log lines 27-38; 6_report.json line 5; results.md line 61
Static IR drop, VSS	worst 0.6275 mV (6.27544e-04 V), average 0.0872 mV, 0.03 %; all VSS shapes connected	same	VSS 0.0 V (GND_NETS_VOLTAGES), same activity	VERIFIED	6_report.log lines 39-50; 6_report.json line 9; results.md line 62
Same analysis re-run	VDD worst 4.54e-04 V, VSS worst 6.28e-04 V, identical	PSM analyze_power_grid	ir.tcl	REPRODUCED 2026-10-06	ir.log lines 14, 33
Dynamic / vectored IR drop	not run	—	—	MISSING	—

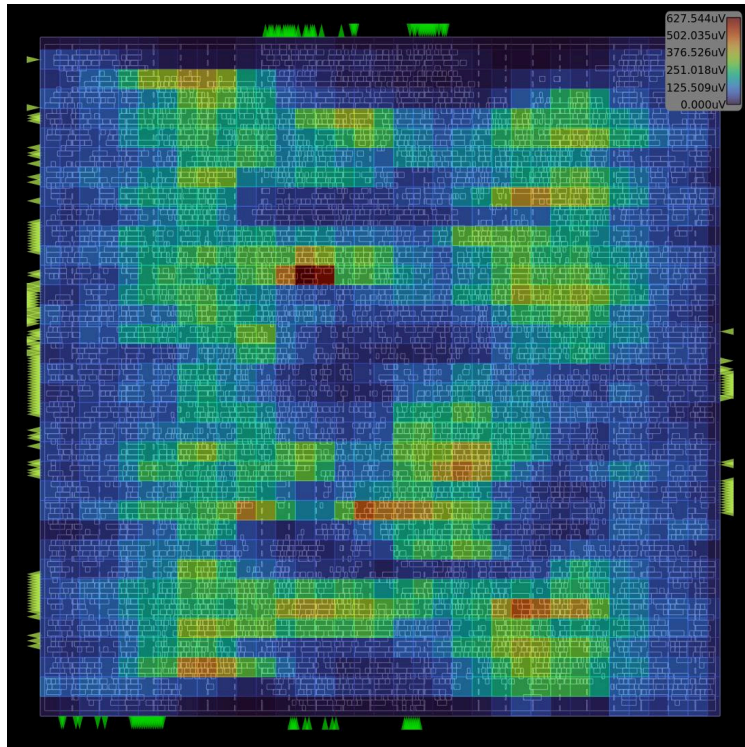


Figure 2: ORFS GUI export `final_ir_drop` (static, default activity). The export does not name the net; its legend maximum 627.544 uV equals the VSS worst-case drop, so it most likely shows VSS (ASSUMPTION). Source [final_ir_drop.webp.png](#) (= `build/pd_sep/reports/sky130hd/orbit_demo/sep/final_ir_drop.webp.png`).

10 Electromigration

Check	Result	Tool + version	Conditions	Status	Source
EM in the flow	NOT RUN: ORFS calls <code>analyze_power_grid</code> without <code>-enable_em</code> ; no EM text in any sep log (grep of <code>build/pd_sep/logs/...</code> and <code>make_pd_sep.out</code> , 2026-10-06)	—	—	MISSING	ORFS <code>final_outputs.tcl</code> lines 39-56; orfs_logs/
Power-grid currents only (scratch, for information)	max segment current VDD 4.05e-04 A, VSS 4.33e-04 A; no current-density limits applied, so no pass/fail	PSM <code>analyze_power_grid -enable_em</code>	same default activity; per-segment reports <code>em_vdd.rpt</code> / <code>em_vss.rpt</code> stay in review scratch	REPRODUCED 2026-10-06 (currents only)	ir.log lines 17-23, 36-42 (max current lines 20, 39)
EM pass/fail (power grid; signal and clock nets)	not run. <code>sky130_fd_sc_hd.tlef</code> (read by the flow, 1_synth.log line 7 ODB-0227) defines <code>DCCURRENTDENSITY</code> / <code>ACCURRENTDENSITY</code> for <code>mcon</code> , <code>via-via4</code> and <code>met1-met5</code> (15 entries); no step compared the PSM currents or signal/clock-net currents with them	—	—	MISSING	<code>sky130_fd_sc_hd.tlef</code> (ORFS image copy; 13 layers and 25 vias, as in the ODB-0227 line) lines 91, 118-119, 135, 159-160, 178, 200-201, 218, 241-242, 260, 282-283

11 Formal property verification (fault-free RTL)

Conditions for every row: harness `orbit_demo_fv.sv` with an independent reference model written from `SPEC.md`; every DUT input free in every cycle; only assumption reset in the first cycle (`c1clear` starts from an unconstrained state, `wrap` from a consistent arbitrary state); parameters fixed LANES = 4, T_THROTTLE = 80, T_STOP = 95, T_RECOVER = 70; Yosys 0.69+154, SBY v0.69, Yices 2.7.0, bitwuzla 0.9.1; FORMAL_TIMEOUT 1800 s. Published run 2026-09-30 00:14 UTC ([formal/summary.md](#) lines 3-4). Re-run 2026-10-06 at 0495cfa with a scratch BUILD. `formal/summary.md` names the RTL only as `rtl`, with no hash; see discrepancy register (09_review_notes).

11.1 Tasks

Task	Property groups	Mode / engine / depth	Assertions / covers	Published (2026-09-30)	Re-run 2026-10-06	Status	Source
thermal	P5, P6 (+4 helpers)	prove, smtbmc yices, k-induction k = 8	15	PASS, 0 s	PASS	REPRODUCED 2026-10-06	formal/summary.md line 28; logs/thermal.log ; make_formal.log
dup	P7	prove, smtbmc yices, k = 6	12	PASS, 0 s	PASS	REPRODUCED 2026-10-06	line 29; logs/dup.log
handshake	P2, P3, P4 + thermal + dup	prove, smtbmc yices, k = 8	41	PASS, 0 s	PASS	REPRODUCED 2026-10-06	line 30; logs/handshake.log
datapath	P1, P7, P8 (+8+8 helpers)	prove, smtbmc yices, k = 6	59	PASS, 3 s	PASS	REPRODUCED 2026-10-06	line 31; logs/datapath.log
clear	P5, P8 from an unconstrained start state (no reset)	prove, smtbmc yices, k = 3	8	PASS, 0 s	PASS	REPRODUCED 2026-10-06	line 32; logs/clear.log
product	P1 lemma $\text{sext32}(\text{signed8}(a) * \text{signed8}(b)) == \text{sign} * (\text{abs}(a) * \text{abs}(b))$, all 2^{16} operand pairs	prove, smtbmc bitwuzla, k = 1	1	PASS, 6 s	PASS	REPRODUCED 2026-10-06	line 33; logs/product.log
thermal_pdr	P5, P6, no helper invariants	prove, abc pdr (unbounded)	11	PASS, 2 s	PASS	REPRODUCED 2026-10-06	line 34; logs/thermal_pdr.log
dup_pdr	P7	prove, abc pdr	12	PASS, 2 s	PASS	REPRODUCED 2026-10-06	line 35; logs/dup_pdr.log
handshake_pdr	P2, P3, P4, no helpers	prove, abc pdr	13	PASS, 7 s	PASS	REPRODUCED 2026-10-06	line 36; logs/handshake_pdr.log
live	P2_live_delivered_s_eventually !f_watch under fairness_s_eventually out_ready	live, aiger suprove (liveness-to-safety)	1	PASS, 25 s; suprove gives no per-property status or trace	PASS, 17 s	REPRODUCED 2026-10-06	line 37; logs/live.log
cover	C01-C14 from reset	cover, smtbmc yices, depth 24	14 covers, all reached at step 3-6	PASS	PASS, same steps	REPRODUCED 2026-10-06	lines 38, 44-59; logs/cover.log
wrap	C20-C22 INT32 wraparound from a consistent arbitrary state	cover, smtbmc yices, depth 8, cover_assert	3 covers reached at step 2; 88 assertions UNKNOWN (checked only along the 3 traces, not proven by this task)	PASS	PASS	REPRODUCED 2026-10-06	lines 39, 60-62, 313-404; logs/wrap.log
datapath_pdr	P1, P8, no helpers (make formal-extra only)	prove, abc pdr	31	PASS, 837 s	PASS, 941 s (15 m 41 s wall)	REPRODUCED 2026-10-06	line 40; logs/datapath_pdr.log ; make_formal-extra.log ; summary_extra.md

Re-run totals: make formal "[formal] all 12 tasks passed", 46.6 s wall, exit 0 ([make_formal.log](#)); per-property status P1-P8 and cover steps identical to the published summary ([formal_summary.md](#) lines 13-62).

11.2 Per-property result

Property	Meaning (short)	Result	Status	Source
P1	every out_fire delivers the reference result per lane, INT32 wraparound	PROVEN unbounded (datapath k = 6, product, datapath_pdr)	REPRODUCED 2026-10-06	formal/summary.md line 15
P2	every accepted in_last gives exactly one out_fire, in order	PROVEN unbounded (handshake, handshake_pdr, live)	REPRODUCED 2026-10-06	line 16
P3	out_valid && !out_ready: held and stable	PROVEN unbounded	REPRODUCED 2026-10-06	line 17
P4	in_ready low in STOP/3, fault, clear_fault, buffer full && !out_ready	PROVEN unbounded; the code-3 branch is vacuous because voted code 3 is unreachable fault-free (P5_never_code3) and mutant m27 survives (section 14); see discrepancy register (09_review_notes)	REPRODUCED 2026-10-06	line 18
P5	voted thermal state follows the SPEC table; reset gives STOP	PROVEN unbounded (P5_shutdown_req vacuous for code 3, as P4)	REPRODUCED 2026-10-06	line 19
P6	throttle: first THROTTLE cycle admits, then alternate	PROVEN unbounded	REPRODUCED 2026-10-06	line 20
P7	fault-free: fault and therm_repair never rise, copies equal	PROVEN unbounded	REPRODUCED 2026-10-06	line 21
P8	clear_fault gives fault = 0, out_valid = 0, storage zero, next beat sums from 0 (also from a latched fault)	PROVEN unbounded	REPRODUCED 2026-10-06	line 22
Induction depth	"closes at k = 2 for thermal, handshake and datapath and at k = 1 for dup (measured 2026-09-29)"; the logs are consistent (third / second "Trying induction" step succeeds) but the measurement itself has no log	—	CLAIM (unverified)	formal/summary.md lines 504-508
Development claims	"a first attempt ran for over 10 minutes at step 4"; the clear task "added after an independent review found ..."	no log	CLAIM (unverified)	lines 499-500, 516-517

SPEC section 6 says phase is "forced to 0 whenever the voted state is not THROTTLE"; the RTL clears it at the next edge. The formal summary documents the difference (lines 525-532) and P6 proves admission is unaffected; see discrepancy register (09_review_notes).

11.3 Vacuity check (19 one-line RTL mutants that the proofs must fail)

Method: make formal-vacuity applies one sed edit to a generated RTL copy (never rtl/) and runs the task that should catch it; CAUGHT = sby FAIL with a counterexample from reset on a property with the expected prefix; ERROR, UNKNOWN or timeout (FORMAL_VAC_TIMEOUT 900 s) count as MISSED ([formal/vacuity.md](#) lines 3-7).

Result	Tool + version	Conditions	Status	Source
19 / 19 CAUGHT (published 2026-09-30 00:13:00:14)	SBY v0.69, smtbmc yices / abc pdr / aiger suprove	14 mutants on the k-induction tasks, 5 on the pdr tasks	VERIFIED	formal/vacuity.md lines 11-29; formal/logs/vacuity_*.log

Result	Tool + version	Conditions	Status	Source
19 / 19 CAUGHT, "[vacuity] all mutants caught", 31.4 s	same	re-run at 0495cfa, scratch BUILD	REPRODUCED 2026-10-06	formal_vacuity.md ; make_formal-vacuity.log
19 / 19 CAUGHT, 31.0 s (second, independent re-run)	same	separate scratch BUILD	REPRODUCED 2026-10-06	make_formal-vacuity_second_run.log

The verdicts are stable; the failing property and step reported for two abc-pdr mutants are not:

Mutant	Published	Re-run A (package formal_rerun)	Re-run B (second run)
zext_product_pdr	P1_out_valid_data_lane1 P8_fresh_sum_from_zero_lane1, step 4	P8_fresh_sum_from_zero_lane2, step 6	P1_out_valid_data_lane3 P8_fresh_sum_from_zero_lane3, step 4
clear_keeps_acc_pdr	P8_clear_zero_acc_a_lane0 P8_clear_zero_acc_b_lane0, step 5	same properties, step 8	same properties, step 7

The other 17 rows (property names and steps) are identical in all three runs. The "step 0" entries for no_backpressure and no_backpressure_pdr cannot be checked from the trimmed published logs; see discrepancy register (09_review_notes).

11.4 Formal limits

Fault-free RTL only (fault detection is section 12); RTL only, no property check on any netlist; single clock; parameters fixed; covers from reset searched to depth 24; the harness depends on Yosys hierconn. Source: [formal/summary.md](#) lines 534-549. Status of any netlist-level property check: MISSING.

12 Fault injection

12.1 Formal fault injection on generated RTL copies

Conditions: [fault/gen_fault_copies.py](#) writes one RTL copy per scenario with one storage element made upsettable ($q \leq q_{\text{next}} \wedge \text{fi_mask}, \text{fi_bit} = \text{\$anyconst}, \text{fi_req} = \text{\$anyseq}$); production RTL is the fault-free twin in [fi_harness.sv](#); at most one upset per trace, any cycle including reset cycles, any lane, any bit; protected scenarios: prove with k-induction $k = 4$ (FAULT_PROVE_DEPTH), cover depth 12; negative scenarios: BMC depth 12; engine smtbmc yices; Yosys 0.69+154, SBY v0.69, Yices 2.7.0. Injection is on RTL copies, not on any netlist. Published run 2026-09-30 00:41-00:44; re-run 2026-10-06 with make BUILD=<scratch> FAULT_PUBLISH= fault (reports/fault not written), "[fault] all checks passed", 3 m 17 s.

Scenario	Injected element	Result	Status	Source
acc_a, acc_b	u_acc_a, u_acc_b, any lane/bit/cycle	prove PASS, 18 assertions, induction closed; cover 8/8	REPRODUCED 2026-10-06	fault/summary.md lines 45-46; fault/logs/acc_a_prove.log ; formal_results.md
res_a, res_b	u_res_a (drives out_data), u_res_b	prove PASS, 18 assertions, induction closed; cover 8/8	REPRODUCED 2026-10-06	lines 47-48; fault/logs/
therm_c0, therm_c1, therm_c2	u_thermal.u_copy0/1/2, any bit/cycle	prove PASS, 16 assertions, induction closed; cover 8/8 (incl. C_upset_copy_code3, C_upset_in_throttle, C_upset_in_stop)	REPRODUCED 2026-10-06	lines 49-51
neg_out_valid_q (unprotected)	out_valid_q	cover 5/5; fire FAIL as expected (D_out_fire_matches_ref, step 1); loss FAIL as expected (D_no_silent_loss, step 3)	REPRODUCED 2026-10-06	line 52; neg_out_valid_q_fire.log , neg_out_valid_q_loss.log
neg_product (unprotected)	one-cycle transient on the shared prod32 of one lane	cover 2/2; fire FAIL as expected (D_out_fire_matches_ref, step 3); wrong out_data transferred with no fault (SDC)	REPRODUCED 2026-10-06	line 53; neg_product_fire.log ; neg_product.diff
Negative control no_compare (assign mismatch = 1'b0;)	acc_a, res_a on the mutant	4 / 4 FAIL as expected (L_stop_in_upset_cycle step 1; D_out_fire_matches_ref step 3)	REPRODUCED 2026-10-06	lines 143-163; negctl_no_compare.diff ; negctl_formal_results.md
Totals	19 scenario tasks, 0 unexpected; 4 negctl tasks, 0 unexpected	—	REPRODUCED 2026-10-06	make_fault.log ; summary.md (re-run)
Hand-run BMC without helpers (about step 8 / frame 10 after about 10 min) and 24-step BMC with helpers (7 min)	no log	—	CLAIM (unverified)	fault/summary.md lines 57-62

12.2 RTL SEU simulation campaign

Conditions: [tb_seu_campaign.v](#), Icarus Verilog 14.0 (devel); two copies of the production RTL behind identical hosts on one random workload; seed=1 tpb=12 tpb_unprot=200 controls=20 prefix=8..127 post=64 drain=24+8; one bit flipped by hierarchical assignment at the falling edge; zero-delay RTL, 10 ns sim clock; host reaction clear_fault 2 cycles after fault. Re-run 2026-10-06: trials.tsv byte-identical to build/fault/campaign/trials.tsv, report byte-identical to the published campaign.md.

Storage group	Result (trials: outcomes)	Status	Source
All	6836 trials, 0 bench errors; 20 controls all MASKED; 6816 injected; all 521 bits, each at least 12 times; 9 / 9 campaign checks PASS	REPRODUCED 2026-10-06	fault/campaign.md lines 12, 36-46; campaign_report.md ; per-bit table campaign_per_bit.tsv
Accumulator pairs (256 bits)	3072: MASKED 3, DETECTED 3069, escapes 0	REPRODUCED 2026-10-06	campaign.md line 5
Result pairs (256 bits)	3072: MASKED 4, DETECTED 3067, TIMING 1, escapes 0	REPRODUCED 2026-10-06	line 6
Detection latency (acc + res)	all 6136 detected trials: fault = 1 one cycle after the upset cycle; in_ready and out_valid already 0 at the end of the upset cycle	REPRODUCED 2026-10-06	line 14
8 acc/res upsets not DETECTED	all had clear_fault in the upset cycle (both copies zeroed); 7 MASKED, 1 TIMING (trial 3836 g_lane[1].u_lane.u_res_a.q[30])	REPRODUCED 2026-10-06	lines 16-17
Thermal triple (6 bits)	72: REPAIRED 72, escapes 0	REPRODUCED 2026-10-06	line 7
Throttle phase (unprotected)	200: MASKED 156, TIMING 44 (all in THROTTLE). Escapes column 0 because TIMING is not counted as an escape although SPEC section 8 lists phase as an expected escape; see discrepancy register (09_review_notes)	REPRODUCED 2026-10-06	lines 8, 25-27
out_valid_q (unprotected)	200: LOST 38, EXTRA 162 (129 duplicate, 33 stale), escapes 200 (100 %)	REPRODUCED 2026-10-06	lines 9, 23-24, 30
fault_q 0 -> 1 (unprotected)	200: DETECTED 200, a false fault stop with no data error	REPRODUCED 2026-10-06	lines 10, 28

Storage group	Result (trials: outcomes)	Status	Source
fault_q 1 -> 0 (SPEC section 8 escape)	not measured: needs two faults, the single-upset campaign cannot reach fault_q = 1 first	MISSING	fault/summary.md lines 254-256
Negative control campaign on no_compare	535 trials, 0 errors; acc 256: MASKED 196, SDC 60; res 256: MASKED 230, SDC 26; escape checks PASS	REPRODUCED 2026-10-06	fault/summary.md lines 230-239; negctl_campaign_report.md

12.3 Post-route gate-level fault campaign (sep netlist)

Check	Result	Tool + version	Conditions	Status	Source
GLS lockstep with injection	"GLS PASS: 30000 cycles, 0 mismatches"; 702 duplicated-copy flips (702 stopped), 234 thermal-copy flips (234 repaired); 518 of 518 redundant bits hit; coverage: 8729 beats, 1037 results, 6 resets, 733 clears; NORMAL 14075 / THROTTLE 1737 / STOP 14182 cycles, code3 0	Icarus Verilog (version not logged in 2026-09-29 run); cell models from cells.lib (tt_025C_1v80) via Yosys read_liberty	tb_gls.v : RTL and routed sep 6_final.v in lockstep, zero-delay, seed 1, one flip per 32 cycles at the falling edge (list in gl_inject.vh); checks stop/repair behaviour and lockstep outputs, no golden-result scoreboard	VERIFIED	gls/gls.log lines 1-5 (= build/pd_sep/gls/sep/gls.log = reports/pdsep/gls.log); pdsep.mk lines 127-137
Same campaign re-run	identical output, 2 m 18 s	Icarus Verilog 14.0 (devel) s20260301-500-g2e81fccb-dirty	same	REPRODUCED 2026-10-06	gls_rerun.log
Netlist-mutant control on the sep netlist	instance_4098_sky130_fd_sc_hd_and2_1 -> sky130_fd_sc_hd_or2_1 (#0 of 184): first MISMATCH at cycle 7, "GLS FAIL: 1148 errors" (the harness observes the netlist)	Icarus 14.0	3000 cycles, seed 1. The log does not name the netlist; "#0 of 184" matches the 184 and2_1 instances of the sep 6_final.v (the base netlist has 189)	REPRODUCED 2026-10-06	gls_mutant.log
Directed bench on the sep netlist	see 05_simulation (S8): TB_ORBIT_DEMO PASS, summary identical to RTL	Icarus 14.0	zero-delay	REPRODUCED 2026-10-06	gls_sky130hd_sep_directed.log

12.4 Not injected

Item	Status	Source
Comparators (lane_mismatch, mismatch), voter, clock, reset, sensor and handshake inputs	MISSING (listed in SPEC section 8, not injected in either flow)	fault/summary.md lines 243-246
Multi-bit upsets, common-mode upsets of both copies	MISSING	same
fault_q 1 -> 0 (needs two faults); thermal voted code 3 (needs two copies upset)	MISSING	lines 254-256; mutation/summary.md (m27)
Shared-product transient in the simulation campaign (only formal neg_product)	MISSING	campaign injects stored bits only
Fault injection with SDF timing (SET width, latching window)	MISSING	all GLS zero-delay
Multi-seed campaign statistics, confidence intervals	MISSING (seed 1 only)	fault/summary.md lines 251-253
Radiation test data, cross-sections, FIT	MISSING ("Nothing here is a radiation test")	line 257

13 Synthesis checks

Check	Result	Tool + version	Conditions	Status	Source
keep_hierarchy in the ORFS (reviewed) synthesis	hierarchy kept: 19 orbit_keep_reg submodules (16 x W=32, 3 x W=2 RESET_VAL=2'10); 521 sky130_fd_sc_hd_dfxtpt_1; 5130 cells, 46866.1984 um^2. Kept by (* keep_hierarchy *) in orbit_keep_reg.v line 14 and SYNTH_KEEP_MODULES = *orbit_keep_reg* (config.mk line 30). Log warning Selection "*orbit_keep_reg*\\$" did not match any module (the second, slang-frontend pattern)	Yosys 0.68+post (ORFS)	ABC speed script, -D 7.2, liberty TT	VERIFIED	synth_stat.txt lines 152-154, 265-269; 1_2_yosys.log lines 1-9
ORFS synthesis structural check	CHECK pass "Found and reported 0 problems" on both orbit_keep_reg variants and orbit_demo	Yosys 0.68+post	—	VERIFIED	synth_check.txt
keep negative control, ORFS (baseline platform, synthesis stage only)	cfgkeep (RTL attribute removed, SYNTH_KEEP_MODULES set): PASS 19/19, 521 FF; nokeep (both removed): FAIL 17/19, 517 FF, thermal u_copy1/u_copy2 0 cells	Yosys 0.68+post, pd/audit_storage.py	baseline pd/sky130hd config, not the sep run	VERIFIED	negctl_cfgkeep_audit.txt ; negctl_nokeep_audit.txt
keep negative control, generic flow	nokeep: 517 FF, 4865 cells, thermal copies MERGED, audit FAIL 8/9; regkeep ((* keep *) on q only): same, does not prevent the merge; NEGATIVE_CONTROL PASS	Yosys 0.69+154	synth/synth_generic.py	REPRODUCED 2026-10-06	negative_control.txt (re-run byte-identical, make_synth.log)

Check	Result	Tool + version	Conditions	Status	Source
Storage audit, routed sep 6_final.v (published)	RESULT: PASS (18/18 checks passed); 6250 leaf cells; 521 dfxtp_1; 9 redundant groups OK; 521 distinct Q nets; 51 clock-leaf nets. Produced by the older pd/audit_storage.py (report 2026-09-29 22:30; script changed in b1da230, 23:47, adding a 19th check)	pd/audit_storage.py (pre-b1da230), liberty build/pd_sep/platform/cells.lib	—	VERIFIED	storage_audit.txt lines 3, 36
Storage audit, same netlist, current script	RESULT: PASS (19/19); added check "clock pins on the clock tree: 0 flip-flops whose clock net does not trace back to port 'clk'" PASS; all other lines identical	pd/audit_storage.py at HEAD	same netlist and liberty	REPRODUCED 2026-10-06	storage_audit_current_script.txt lines 35, 37
Storage audit, generic netlist	STORAGE_AUDIT PASS: 9/9 groups, 521 FF bits, 3 unprotected (fault_q, out_valid_q, u_thermal.phase); flat view identical; self-test: 12 defect classes all detected	Yosys 0.69+154, scripts/storage_audit.py	generic netlist build/synth/orbit_demo_synth.json	REPRODUCED 2026-10-06	synth/storage_audit.txt ; storage_audit_flat.txt ; audit_selftest.log
Generic synthesis structural checks	SYNTH_GENERIC PASS: 4899 cells = 4373 generic logic + 521 FF + 5 \$scopeinfo; no latches; no warnings	Yosys 0.69+154	synth -flatten with kept instances	REPRODUCED 2026-10-06	synth_counts.txt (re-run byte-identical)
Equivalence RTL vs generic netlist	EQUIV PASS: 47 register ports cut and matched by name; tasks lane0-lane3, ctrl PASS; mutants mut_throttle (T_THROTTLE 80 -> 81) and mut_xext FAIL as required	SymbiYosys, smtbmc bitwuzla, BMC depth 1 per partition (register-correspondence combinational check)	gold RTL vs build/synth/orbit_demo_synth.v (generic cells, not sky130)	REPRODUCED 2026-10-06	equiv_result.txt ; re-run (44 s vs 40 s wall)
GLS of the generic netlist	TB_ORBIT_DEMO PASS, summary identical to the RTL run (14 scenarios, 133601 cycles, 0 errors)	Icarus 14.0, Yosys simcells/simlib	zero-delay, seed 1	REPRODUCED 2026-10-06	gls_result.txt ; make_synth.log
make synth overall	"SYNTH PASS: generic synthesis + checks, storage audit (+ self-test), keep_hierarchy negative control, equivalence to RTL, gate-level simulation", 2 m 0.2 s	Yosys 0.69+154	scratch BUILD	REPRODUCED 2026-10-06	make_synth.log
RTL vs sky130-mapped netlist (l2_yosys.v / l_synth_1ec.v) formal equivalence	not run; only zero-delay GLS links RTL to the sky130 netlists (12.3)	—	—	MISSING	—
Brief p.4 "4,356 generic logic cells"	not reproduced: 4373 native, 4384 YoWASP Yosys 0.69; no script given	—	—	CLAIM (unverified)	synth/summary.md lines 12, 21; yowasp_compare.txt

Two different storage audits exist (generic [scripts/storage_audit.py](#), "9/9 groups"; sky130 [pd/audit_storage.py](#), "18/18" or "19/19 checks"), and the published sep reports (results.md line 24, checks.json, summary.md) still quote the superseded 18/18; see discrepancy register (09_review_notes).

14 Mutation testing of the verification suite (context for formal and fault)

Check	Result	Tool + version	Conditions	Status	Source
Mutation score	35/40 valid mutants killed (87.5 %); 35/39 (89.7 %) excluding m21, argued equivalent	scripts/mutation_test.py ; targets lint, sim, formal-quick, fault-quick, synth	40 RTL mutants, baseline m00 passes all targets; generated 2026-09-30; not re-run	VERIFIED	mutation/summary.md lines 5-52
Survivors	m07 result_from_b, m20 out_valid_no_fault, m22 fault_not_sticky, m27 shutdown_only_stop (real test gaps), m21 out_fire_unmasked (equivalent); all also pass the full formal set. Missing checks named: fault -> !out_valid && !in_ready and fault stickiness under a fault_q upset; out_data equal to reference while out_valid low; therm_state == 3 -> shutdown_req && !in_ready	evidence bench tb_mutation_gaps.v FAILs on each survivor, PASS on unmodified RTL	—	VERIFIED	mutation/summary.md lines 72-140

15 ECC standalone block (not in the layout)

[rtl/ecc/orbit_secdded72.v](#) (Hsiao (72,64) SECDED encoder/decoder) and [rtl/ecc/orbit_ecc_bank.v](#) (16 rows x 2 interleaved codewords, scrubber, error log) are not instantiated by orbit_demo, are not in \$(RTL), and were not read by the ORFS synthesis of the reviewed layout ([l1_yosys_canonicalize.log](#) lines 7-11 read only the four orbit_demo RTL files; [mk/ecc.mk](#) line 3 "NOT part of orbit_demo"). Physical checks (timing, area, power, DRC, LVS, IR, EM) of ECC: N/A for this layout (block not placed); MISSING if ECC is to be integrated.

Conditions: published run [ecc/results.txt](#) (= reports/ecc/results.txt); full make BUILD=<scratch> ecc re-run 2026-10-06 at 0495cfa: "ECC: PASS", exit 0, 20 m 41 s; summary identical to the published results except run times. Tools Yosys 0.69+154, SBY, smtbmc bitwuzla/yices, Icarus 14.0, Verilator 5.053.

Check	Result	Tool	Conditions	Status	Source
Lint	Verilator -Wall and Icarus -Wall clean	Verilator 5.053, Icarus 14.0	--top-module orbit_ecc_bank	REPRODUCED 2026-10-06	ecc/results.txt line 14; summary.txt

Check	Result	Tool	Conditions	Status	Source
H matrix	Hsiao construction: 72 distinct odd-weight columns, 26 ones per data row	rtl/ecc/gen_hsiao72.py -- check	—	REPRODUCED 2026-10-06	results.txt line 13; hsiao.log
Codec simulation	seed 1: 20003 no-error words; 21600 single-bit errors corrected (300 words x 72); 30672 double-bit errors detected (12 x 2556 pairs); 19189 triple-bit: 8417 UE, 10772 miscorrected; errors 0; PASS	Icarus 14.0	reference H built from the construction rule	REPRODUCED 2026-10-06	sim_codec_seed1.log
Bank simulation	seeds 1 and 2, 15098 cycles each (15000 random + directed), errors 0; seed 1: 3462 writes / 4971 reads, 270 CE / 14 UE reads, 1075 scrub repairs	Icarus 14.0	DEPTH 16, INTERLEAVE 2, CNT_W 16 plus a CNT_W 3 copy; upsets by hierarchical writes, never a third error per codeword	REPRODUCED 2026-10-06	sim_bank_seed1.log , sim_bank_seed2.log
Simulation negative controls	7 / 7 mutants killed (bank_cnt_wrap, bank_log_prio, bank_no_interleave, bank_no_writeback, bank_read_way_swap, bank_scrub_on_read, codec_even_col)	Icarus 14.0	exact text substitution, bank bench 3000 cycles	REPRODUCED 2026-10-06	results.txt lines 18-24
Codec proof	PASS: no error unchanged; any single error corrected (ce=1); any double error ue=1, no flip; encoder linear	SBY prove depth 1, smtbmc bitwuzla	combinational, all inputs free (complete)	REPRODUCED 2026-10-06	results.txt line 8; formal_codec_orbit_secdded72_codec.log
Bank proofs clean / sec / ded	PASS by k-induction (41 s / 135 s / 89 s published)	SBY prove depth 2, smtbmc bitwuzla	DEPTH 16, INTERLEAVE 2, CNT_W 3; decoder replaced by an abstraction constrained to the codec contract; at most one upset event per edge (1 bit or 2 adjacent columns)	REPRODUCED 2026-10-06	results.txt lines 1, 4, 7; formal_bank_*.log
Covers	cover: c_scrub_two step 2, c_ce_read_adj 3, c_saturate 6, c_bound_tight 17; cover_ded: c_ue_read step 4	SBY cover depth 24, smtbmc yices	—	REPRODUCED 2026-10-06	results.txt lines 2-3
End-to-end with the real decoder	e2e_clean PASS (BMC 5 steps), e2e_upset PASS (BMC 4 steps); bounded only; U4 scrub bound not reached at this depth	SBY bmc, bitwuzla	DEPTH 2 (chparam)	REPRODUCED 2026-10-06	results.txt lines 5-6
Unbounded end-to-end statement (real encoder + decoder + bank)	rests on a written composition argument, not machine-checked	—	—	CLAIM (unverified)	rtl/ecc/README.md lines 186-206
Formal negative controls	4 / 4 FAIL as expected (il1_neg, codec_even_col, bank_no_writeback, bank_read_way_swap)	SBY	—	REPRODUCED 2026-10-06	results.txt lines 9-12
sec_bmc positive control, published log	incomplete run of an older harness: steps 0-3 without failure, terminated at 600 s during step 4; neither PASS nor FAIL; not listed in results.txt	SBY bmc depth 6, bitwuzla	DEPTH 16, INTERLEAVE 2, CNT_W 3; harness before commit e80a7b2	VERIFIED (as an incomplete run)	formal_sanity_sec_bmc_production_timeout.log
sec_bmc positive control, re-run with the current harness	PASS at depth 6 (steps 0-5), 2672 s	SBY bmc depth 6, bitwuzla	same parameters, harness at 0495cfa	REPRODUCED 2026-10-06	sanity_sec_bmc_result.txt ; sanity_sec_bmc_production_pass_trimmed.log
Generic synthesis	enc 154 cells, 0 FF, depth 6; dec 431 cells, 0 FF, depth 18; bank 10690 cells, 2414 FF (expected 2414), depth 48; no latches	Yosys 0.69+154	generic cells, no liberty	REPRODUCED 2026-10-06	results.txt lines 25-27; ecc/synth_stat.txt

The stale sec_bmc timeout log and the orphan [sim_bank_quick.log](#) (= reports/ecc/logs/sim_bank_quick.log) are flagged; see discrepancy register (09_review_notes).

16 Not done / missing

Item	Why it matters here	What would be needed	Status
Sign-off DRC (FEOL, density, latch-up), Magic DRC	KLayout ORFS deck runs BEOL + OFFGRID only; no fill	Magic DRC or the open_pdk KLayout sign-off deck on a filled GDS	MISSING
Metal fill and density checks	no fill inserted; sky130_fd_sc_hd.tlef (read by the flow, 1_synth.log ODB-0227) defines MAXIMUMDENSITY 70 / DENSITYCHECKWINDOW 700 700 (met1-met4), but no step compared the metal density with it	fill generation, then a density check against these values or a sign-off deck	MISSING
Independent LVS	KLayout LVS reference comes from the same 6_final.odb	Magic extraction + Netgen against 6_final.v or the synthesized netlist	MISSING
Antenna check outside OpenROAD	only the OpenROAD ANT checker ran	Magic / KLayout antenna rules	MISSING
RTL vs sky130 netlist formal equivalence	kepler-formal covers synth -> final only; RTL -> generic proven, RTL -> sky130 only simulated	eqy or equiv_cut.yf flow with sky130 cell models on 1_2_yosys.v / 6_final.v	MISSING

Item	Why it matters here	What would be needed	Status
kepler-formal, OpenROAD, ORFS, PDK versions	not recorded in any log	log image digest, ORFS/OpenROAD commit, open_pdk version per run	MISSING
RC corners, OCV / derates, clock uncertainty, I/O budgets	timing uses nominal RC and placeholder I/O delays	min/max RCX rules, uncertainty/derate spec, interface budget	MISSING
Additional liberty corners; timing closure at SS	SS fails by 6.131 ns with 516 endpoints and 37/12 slew/cap violations	more corners; MCMC optimisation, or a decision on operating corner and period	MISSING
Demonstrator speed / PVT requirement	no requirement to judge corner results against	stated target frequency and PVT range	MISSING
Post-layout SPICE; SDF-annotated GLS	all GLS zero-delay; no transistor-level simulation	SDF from OpenSTA with 6_final.spf; timing GLS; SPICE of critical paths if needed	MISSING
Activity-based and corner power; dynamic IR	power and IR use default activity at TT	VCD/SAIF from a workload; read in OpenSTA; vectored PSM	MISSING
Electromigration	not run; sky130_fd_sc_hd.tlef (read by the flow, 1_synth.log ODB-0227) defines DCCURRENTDENSITY / ACCURRENTDENSITY per via and metal layer, but no step compared the PSM currents (section 10) or signal/clock-net currents with them	the comparison, per layer and via, for power, signal and clock nets	MISSING
Fault injection gaps	comparators, voter, clock, reset, inputs, MBU, common mode, fault_q 1 -> 0, thermal code 3, product in campaign, netlist/SDF injection, multi-seed	see 12.4; double-fault scenarios; netlist-level campaign	MISSING
Property checks the mutation run asks for	m07, m20, m22, m27 survive	assertions listed in section 14	MISSING
Formal on a netlist	all properties checked on RTL only	rerun orbit_demo_fv.sv on a netlist	MISSING
Regenerated published sep storage audit	published reports quote 18/18 from a superseded script	re-run make pd-sep-report and republish	MISSING (re-run only in this package)
Make target for sep multi-corner STA	results exist only as a hand run	pd-sep-corners target in mk/pdsep.mk publishing to reports/pdsep/	MISSING
Counterexample / cover VCDs, per-trial campaign log	only in gitignored build/	copy build/formal, build/fault traces and trials.tsv	MISSING (not in package)
LEC netlists, make pd-sep console log, 5_1_grt.json, antenna logs, LVS database and reference CDL, 6_final.odt	were only in gitignored build/ (D-99)	—	resolved in package: additional_evidence/ , 6_final.odt.gz
ECC integration and physical data	ECC is standalone	integration spec, RTL, layout run	N/A for this layout
Pad ring, IO cells, ESD, package checks	the layout is a core block: 0 padcells (6_report.json line 60)	—	N/A (no pad ring, no package)
Analog / mixed-signal verification	no analog blocks	—	N/A (digital only)
Silicon measurement	not fabricated	—	N/A (no silicon)